Chapter 10 Rates and Projectile Motion - Homework Answers

Chapter revision | Warm-up

1 **a** $v = 2t - 6$, $a = 2$ **b** $v = 0$ when $t = 3$, at $x = -4$ **c** $x(0) = 5$, $x(3) = -4$, $x(4) = -3$, so the distance is $9 + 1 = 10$ m. Average velocity $= \dfrac{-3 - 5}{4} = -2$ m/s; average speed $= \dfrac{10}{4} = 2.5$ m/s.

2 $\dot{x} = 3t^2 - 4t + 3$, $x = t^3 - 2t^2 + 3t$. For $\dot{x} = 0$, $\Delta = 16 - 36 < 0$, and $\dot{x}$ is a positive definite quadratic, so $\dot{x} > 0$ for all t: never stationary.

3 $\dot{x} = 20 - 10t$, $x = 20t - 5t^2$. Greatest height 20 m when $t = 2$. Back on the ground when $t = 4$, with $\dot{x} = -20$, so the speed is 20 m/s.

4 $40\cos 30° = 20\sqrt{3}$, $40\sin 30° = 20$; $\theta \doteq 37°$; $\sqrt{36 + 64} = 10$

5 $\dfrac{dA}{dt} = 2\pi r \times 3 = 24\pi$ when $r = 4$

Chapter 10 Rates and Projectile Motion - Homework Answers

Lesson 9E | Related rates

Lesson Examples

C1 **Tutor demo. a** $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$ **b** $4\pi \times 25 \times 2 = 200\pi$ mm³/s **c** $V = 36\pi$ gives $r^3 = 27$, $r = 3$; then $72\pi = 4\pi \times 9 \times \dfrac{dr}{dt}$, so $\dfrac{dr}{dt} = 2$ mm/s.

Your turn. $\dfrac{dA}{dt} = 2x \times 0.2 = 0.4x$. **a** 1.6 m²/s **b** $0.4x = 2$, so $x = 5$ m.

C2 **Tutor demo. a** $V = \frac{1}{3}\pi\left(\frac{1}{2}h\right)^2 h = \frac{\pi}{12}h^3$ **b** $\dfrac{dV}{dt} = \frac{\pi}{4}h^2 \dfrac{dh}{dt}$, so $12 = 4\pi\dfrac{dh}{dt}$ and $\dfrac{dh}{dt} = \dfrac{3}{\pi} \doteq 0.95$ cm/s.

Your turn. $V = \frac{1}{3}\pi h^3$, $\dfrac{dV}{dt} = \pi h^2 \dfrac{dh}{dt}$, so $9\pi = 9\pi\dfrac{dh}{dt}$ and the depth increases at 1 cm/s.

C3 **Tutor demo. a** $x = 2\cot\theta$, so $\dfrac{dx}{d\theta} = -2\csc^2\theta = -\dfrac{2}{\sin^2\theta}$. **b** $\dfrac{dx}{dt} = -600$, so $\dfrac{d\theta}{dt} = -\dfrac{\sin^2\theta}{2} \times (-600) = 300\sin^2\theta = 75$ radians per hour, which is $75 \times \dfrac{180}{\pi} \div 3600 \doteq 1.19°$ per second.

Your turn. $x = \dfrac{80}{\tan\theta}$, $\dfrac{dx}{d\theta} = -\dfrac{80}{\sin^2\theta}$, $\dfrac{dx}{dt} = -4$, so $\dfrac{d\theta}{dt} = \dfrac{\sin^2\theta}{20} = 0.0125 \doteq 0.01$ rad/s $\doteq 0.72°$ per second (increasing).

Textbook Homework

1 **b** 1 m²/s **c** 7 metres **d** 9 m²

2 **a** $A = \frac{1}{2}\ell^2$ **c** (i) 5 m²/s (ii) 3 m²/s **d** 34 metres

3 **a** 15.1 m³/s **b** 30.2 m²/s

4 **b** $\dfrac{2}{9\pi}$ cm/s **c** $r = \dfrac{10}{\sqrt{\pi}}$ cm, $V = \dfrac{4000}{3\sqrt{\pi}}$ cm³

5 **a** $V = \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$, which is the volume of a hemisphere of radius r. **b** $\dfrac{1}{10\pi}$ cm/s

6 **b** 325 radians per hour, about 5 degrees per second

7 $\frac{1}{2}\sin^2 15° \doteq 0.0335$ radians per minute, about 2 degrees per minute

8 **a** $V = 100h^2x$ cm³ **b** The surface is $2h$ cm by $100x$ cm, so the water lost each day is $0.1 \times 200xh = 20xh$ cm³. Since $\dfrac{dV}{dh} = 200xh$, the depth of the water changes at a constant 0.1 cm/day (it falls by 0.1 cm each day).

10 **a** $\dfrac{dx}{dt} = -2\sqrt{1-x^2}$ **b** -2 m/s. As the point crosses the y-axis it is travelling horizontally (to the left) at a speed of 2 m/s.

11 **a** $\dfrac{2CV^2}{L^2}$ m/s² **c** As L decreases, the speed passing the truck increases, so the driver should wait as long as possible before beginning to accelerate. A similar result is obtained if the distance between car and truck is increased. Optimally, the driver should allow both L to decrease and C to increase. **d** 950 metres

12 **b** This is just two applications of the chain rule. **d** 6

13 **c** $x = h = 50\left(\sqrt{3}+1\right)$ metres **d** $\dfrac{dx}{dt} = -\dfrac{500}{9}$ m/s, so the speed is 200 km/h.

Chapter 10 Rates and Projectile Motion - Homework Answers

Lesson 9E | Related rates

Past-Paper Homework - Worked Solutions

PP 1 **Caringbah High School, 2020, Higher School Certificate Trial Examination, Mathematics Extension 1, Q2**

$V = \frac{4}{3}\pi r^3$, so $\dfrac{dV}{dr} = 4\pi r^2$. Then $\dfrac{dr}{dt} = \dfrac{dV}{dt} \div \dfrac{dV}{dr} = \dfrac{200\pi}{4\pi \times 10^2} = 0.5$ cm s^{-1}.

Answer (B).

PP 2 **Sydney Girls High School, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q11(d)**

$A = \pi r^2$, so $\dfrac{dA}{dr} = 2\pi r$ and

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} = 2\pi \times 8 \times 2 = 32\pi.$$

The area is increasing at 32π cm^2/s (about 100.5 cm^2/s).

PP 3 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q13(e)**

Let x m be the man's distance from the closest point of the path, and θ the angle in the diagram. Then $\tan\theta = \dfrac{x}{6}$, and $\dfrac{dx}{dt} = 1.2$. Differentiating with respect to t:

$$\sec^2\theta \frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt}, \quad \text{so} \quad \frac{d\theta}{dt} = \frac{1}{6}\cos^2\theta \frac{dx}{dt}.$$

When $x = 8$, the distance from the searchlight to the man is $\sqrt{6^2 + 8^2} = 10$, so $\cos\theta = \dfrac{6}{10} = \dfrac{3}{5}$. Hence

$$\frac{d\theta}{dt} = \frac{1}{6} \times \frac{9}{25} \times 1.2 = \frac{9}{125} = 0.072.$$

The searchlight is rotating at $\dfrac{9}{125}$ radians per second.

PP 4 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q14(c)**

The full cone has radius 3 m and height 5 m.

By similar triangles $\dfrac{r}{h} = \dfrac{3}{5}$, so $r = \dfrac{3h}{5}$. Then

$$V = \frac{1}{3}\pi\left(\frac{3h}{5}\right)^2 h = \frac{3\pi h^3}{25}, \qquad \frac{dV}{dh} = \frac{9\pi h^2}{25}.$$

The water leaks out at 1.5 L/min, so $\dfrac{dV}{dt} = -0.0015$ m^3/min. Then

$$\frac{dh}{dt} = \frac{dV}{dt} \div \frac{dV}{dh} = -0.0015 \times \frac{25}{9\pi h^2}.$$

When $h = 1$: $\dfrac{dh}{dt} = -\dfrac{1}{240\pi} \approx -0.00133$. The water level is falling at $\dfrac{1}{240\pi} \approx 0.00133$ m/min, that is, about 1.3 mm per minute.

PP 5 **North Sydney Girls High School, 2023, HSC Trial Examination, Mathematics Extension 1, Q13(d)**

(i) Take a vertical cross-section through O and the centre of the table. Let the shadow have radius ρ m. The lamp is h m above the table top and $h+1$ m above the ground. The two right triangles with vertex O are similar, so

$$\frac{\rho}{h+1} = \frac{3}{h}, \quad \text{so} \quad \rho = \frac{3(h+1)}{h} = 3\left(1 + \frac{1}{h}\right).$$

Hence $A = \pi\rho^2 = 9\pi\left(1 + \dfrac{1}{h}\right)^2$.

(ii) $\dfrac{dA}{dh} = 9\pi \times 2\left(1 + \dfrac{1}{h}\right) \times \left(-\dfrac{1}{h^2}\right) = -\dfrac{18\pi}{h^2}\left(1 + \dfrac{1}{h}\right)$. When $h = 3$:

$$\frac{dA}{dh} = -\frac{18\pi}{9} \times \frac{4}{3} = -\frac{8\pi}{3}.$$

The lamp is being lowered, so $\dfrac{dh}{dt} = -\dfrac{3}{100}$. Then

$$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt} = -\frac{8\pi}{3} \times \left(-\frac{3}{100}\right) = \frac{2\pi}{25}.$$

The area of the shadow is increasing at $\dfrac{2\pi}{25}$ m^2/s.

PP 6 **North Sydney Boys High School, 2022, HSC Course Assessment Task 3-Trial Examination, Mathematics Extension 1, Q12(b)**

(i) The centre of the sphere lies on the axis of the cone, R below the vertex, so it is $|2-R|$ from the plane of the base. A point on the rim of the base is on the sphere, so by Pythagoras

$$r^2+(2-R)^2=R^2, \quad \text{so} \quad r^2=4R-4.$$

For the slant height, $l^2=r^2+2^2=4R$. Hence

$$A=\pi rl=\pi\sqrt{4R-4}\,\sqrt{4R}=4\pi\sqrt{(R-1)R}=4\pi\sqrt{R^2-R}.$$

(ii) The volume of the sphere is $V=\frac{4}{3}\pi R^3$, so $\dfrac{dV}{dR}=4\pi R^2$.

Then $\dfrac{dR}{dt}=\dfrac{8}{4\pi R^2}=\dfrac{2}{\pi R^2}$. When $R=2$, $\dfrac{dR}{dt}=\dfrac{1}{2\pi}$.

Also $A=4\pi\left(R^2-R\right)^{\frac{1}{2}}$, so $\dfrac{dA}{dR}=4\pi\times\dfrac{1}{2}\left(R^2-R\right)^{-\frac{1}{2}}(2R-1)=\dfrac{2\pi(2R-1)}{\sqrt{R^2-R}}$.

When $R=2$, $\dfrac{dA}{dR}=\dfrac{6\pi}{\sqrt{2}}=3\sqrt{2}\,\pi$. Hence

$$\frac{dA}{dt}=\frac{dA}{dR}\times\frac{dR}{dt}=3\sqrt{2}\,\pi\times\frac{1}{2\pi}=\frac{3\sqrt{2}}{2}.$$

The curved surface area is increasing at $\dfrac{3\sqrt{2}}{2}$ units^2/sec.

Chapter 10 Rates and Projectile Motion - Homework Answers

Lesson 10A | Projectile motion: the time equations

Lesson Examples

C1 **Tutor demo. a** (i) $\underset{\sim}{v} = 10\sqrt{3}\,\underset{\sim}{i} + 10\underset{\sim}{j}$ (ii) $\underset{\sim}{v} = 10\sqrt{2}\,\underset{\sim}{i} - 10\sqrt{2}\,\underset{\sim}{j}$ **b** (i) $v = 10$ m/s, $\theta = \tan^{-1}\frac{4}{3} \doteq 53°$ (ii) $v = 13$ m/s, $\theta = -\tan^{-1}\frac{5}{12} \doteq -23°$
Your turn. a (i) $8\underset{\sim}{i} + 8\sqrt{3}\,\underset{\sim}{j}$ (ii) $8\sqrt{3}\,\underset{\sim}{i} - 8\underset{\sim}{j}$ **b** (i) 13 m/s at $\tan^{-1}\frac{12}{5} \doteq 67°$ (ii) $7\sqrt{2}$ m/s at $-45°$

C2 **Tutor demo. a** $\ddot{x} = 0$, $\dot{x} = 15\sqrt{3}$, $x = 15\sqrt{3}\,t$; $\ddot{y} = -10$, $\dot{y} = -10t + 15$, $y = -5t^2 + 15t$. **b** $\dot{y} = 0$ at $t = 1.5$ s, height 11.25 m above the platform. **c** $y = -20$ gives $t^2 - 3t - 4 = 0$, so $t = 4$ s. Range $60\sqrt{3} \doteq 103.9$ m. At impact $\dot{x} = 15\sqrt{3}$, $\dot{y} = -25$, so $v = \sqrt{1300} = 10\sqrt{13} \doteq 36.1$ m/s at $\tan^{-1}\dfrac{25}{15\sqrt{3}} \doteq 44°$ below the horizontal.
Your turn. $\dot{x} = 10\sqrt{3}$, $x = 10\sqrt{3}\,t$, $\dot{y} = -10t + 10$, $y = -5t^2 + 10t$. Greatest height 5 m (at $t = 1$); flight time 2 s; range $20\sqrt{3} \doteq 34.6$ m.

C3 **Tutor demo. a** $x = 50t\cos\alpha = 200$ gives $t = \dfrac{4}{\cos\alpha}$. Then $y = -80\sec^2\alpha + 200\tan\alpha = 40$, so $2\tan^2\alpha - 5\tan\alpha + 3 = 0$, $\tan\alpha = 1$ or $\frac{3}{2}$: $\alpha = 45°$ or $\alpha \doteq 56°19'$. **b** $t = 4\sqrt{2} \doteq 5.66$ s, or $t = 2\sqrt{13} \doteq 7.21$ s.
Your turn. $t = \dfrac{2}{\cos\alpha}$, and $-20\sec^2\alpha + 80\tan\alpha = 40$ gives $\tan^2\alpha - 4\tan\alpha + 3 = 0$, so $\alpha = 45°$ or $\tan^{-1}3 \doteq 71°34'$.

Textbook Homework

1 **b** $x = 30t$, $y = -5t^2 + 30t$ **c** $t = 6$ seconds **d** 180 m **e** $t = 3$ seconds **f** 45 m

2 **b** $x = 20t\sqrt{3}$, $y = -5t^2 + 20t$ **c** (i) $t = 4$ seconds (ii) $80\sqrt{3}$ m (iii) 20 m

3 **b** $x = 10t$, $y = -5t^2 + 10t\sqrt{3}$ **c** (i) 15.9 m (ii) 12.4 m/s

4 **b** $x = 36t$, $y = -5t^2 + 48t$ **c** 146.5 m

5 **a** $\underset{\sim}{v} = 8\underset{\sim}{i} + (6 - 10t)\underset{\sim}{j}$ **b** $\underset{\sim}{r} = (8t)\underset{\sim}{i} + \left(6t - 5t^2\right)\underset{\sim}{j}$ **c** (i) 10 m/s (ii) $16\underset{\sim}{i} - 8\underset{\sim}{j}$ (iii) $4.8\underset{\sim}{i} + 1.8\underset{\sim}{j}$

6 **a** $\dot{x} = 40$, $\dot{y} = 25$

7 **a** $\underset{\sim}{v} = 4\underset{\sim}{i} + 4\underset{\sim}{j}$ **b** $2\underset{\sim}{i} + 0.75\underset{\sim}{j}$

9 **b** $x = 6t\sqrt{3}$, $y = -5t^2 - 6t$ **c** 1 second **d** 10.4 m

10 **a** 1.19 seconds **b** 5.03 m

11 26 m/s

13 They collide 1 second after P_2 is projected (at $60\underset{\sim}{i} + 45\underset{\sim}{j}$).

14 **c** $\theta = 45°$ or $\theta \doteq 81°52'$

15 **b** (i) $\sqrt{5}$ s and $\sqrt{85}$ s (ii) $20\sqrt{5} \doteq 44.7$ m/s in both cases, at angles of $0°$ and $76.0°$ (below) to the horizontal.

18 **c** $\theta = 60°$

Chapter 10 Rates and Projectile Motion - Homework Answers

Lesson 10A | Projectile motion: the time equations

Past-Paper Homework - Worked Solutions

PP 1 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q7**

Take the origin at O, the foot of the tower. Then $\ddot{y} = -g$. The ball is thrown horizontally, so $\dot{y} = 0$ and $y = 90$ when $t = 0$. Integrating twice, $\dot{y} = -gt$ and $y = 90 - \frac{1}{2}gt^2$.

It strikes the ground when $y = 0$, so $t^2 = \dfrac{180}{g}$ and $t = \sqrt{\dfrac{180}{g}} = \sqrt{\dfrac{36 \times 5}{g}} = 6\sqrt{\dfrac{5}{g}}$ seconds. The horizontal speed of 50 ms^{-1} does not affect the time. **Answer (C).**

PP 2 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(e)**

(i) Differentiating, $\dot{x} = 30\sqrt{3}$ and $\dot{y} = 30 - 10t$. When $t = 0$, $\dot{x} = 30\sqrt{3}$ and $\dot{y} = 30$. If θ is the angle of projection, then

$$\tan\theta = \frac{\dot{y}}{\dot{x}} = \frac{30}{30\sqrt{3}} = \frac{1}{\sqrt{3}}, \quad \text{so} \quad \theta = 30^\circ.$$

(ii) The initial speed V satisfies $V^2 = \dot{x}^2 + \dot{y}^2 = \left(30\sqrt{3}\right)^2 + 30^2 = 3600$, so $V = 60$ m/s.

PP 3 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(d)(i)(ii)**

(i) Resolving the initial velocity, $\underset{\sim}{u} = 30\cos\alpha^\circ\, \underset{\sim}{i} + 30\sin\alpha^\circ\, \underset{\sim}{j}$.

(ii) In components, $\ddot{x} = 0$ and $\ddot{y} = -10$. Integrating, $\underset{\sim}{v} = \underset{\sim}{c} - 10t\, \underset{\sim}{j}$ for some constant vector $\underset{\sim}{c}$. When $t = 0$, $\underset{\sim}{v} = \underset{\sim}{u}$, so $\underset{\sim}{c} = \underset{\sim}{u}$ and

$$\underset{\sim}{v} = 30\cos\alpha^\circ\, \underset{\sim}{i} + (30\sin\alpha^\circ - 10t)\underset{\sim}{j}.$$

Integrating again, $\underset{\sim}{r} = 30t\cos\alpha^\circ\, \underset{\sim}{i} + \left(30t\sin\alpha^\circ - 5t^2\right)\underset{\sim}{j} + \underset{\sim}{d}$ for some constant vector $\underset{\sim}{d}$. The origin is on the ground below the point of projection, so when $t = 0$, $\underset{\sim}{r} = 20\underset{\sim}{j}$. Hence $\underset{\sim}{d} = 20\underset{\sim}{j}$ and

$$\underset{\sim}{r} = 30t\cos\alpha^\circ\, \underset{\sim}{i} + \left(30t\sin\alpha^\circ - 5t^2 + 20\right)\underset{\sim}{j}.$$

PP 4 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q12(d)**

(i) He lands on the top of the pad, where $y = 1$. So $5\left(1 + t - t^2\right) = 1$, that is, $5t^2 - 5t - 4 = 0$. Taking the positive root,

$$t = \frac{5 + \sqrt{105}}{10} \approx 1.5247,$$

so the time of flight is approximately 1.52 seconds.

(ii) At this time, $x = 5\sqrt{3} \times \dfrac{5 + \sqrt{105}}{10} \approx 13.2042$. The pad should be centred 13.20 m from O.

(iii) Differentiating, $\dot{x} = 5\sqrt{3}$ and $\dot{y} = 5(1 - 2t)$. At impact, $1 - 2t = -\dfrac{\sqrt{105}}{5}$, so $\dot{y} = -\sqrt{105}$. Then

$$v^2 = \dot{x}^2 + \dot{y}^2 = 75 + 105 = 180, \quad \text{so} \quad v = 6\sqrt{5} \approx 13.42.$$

His impact speed is about 13 m/s.

(iv) At the highest point $\dot{y} = 0$, so $t = \dfrac{1}{2}$. Then $y = 5\left(1 + \dfrac{1}{2} - \dfrac{1}{4}\right) = \dfrac{25}{4} = 6.25$ m. He misses the ceiling by $8 - 6.25 = 1.75$ m.

PP 5 **Knox Grammar School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(b)**

(i) Since $\tan\alpha = \dfrac{8}{15}$, the velocity triangle at B has sides 8, 15 and $\sqrt{8^2 + 15^2} = 17$. So $\sin\alpha = \dfrac{8}{17}$ and $\cos\alpha = \dfrac{15}{17}$. The firework has not reached its maximum height, so it is still rising, and at B

$$\dot{x} = 34 \times \frac{15}{17} = 30, \qquad \dot{y} = 34 \times \frac{8}{17} = 16.$$

Differentiating $\underset{\sim}{r}$, $\dot{x} = V\cos\theta$ and $\dot{y} = V\sin\theta - 10t$. At $t = 2$, $V\sin\theta - 20 = 16$, so $V\sin\theta = 36$.

(ii) $\dot{x}$ is constant, so $V\cos\theta = 30$. Dividing, $\tan\theta = \dfrac{36}{30} = \dfrac{6}{5}$, so $\theta \approx 50^\circ 12'$. Squaring and adding,

$$V^2 = 36^2 + 30^2 = 2196, \quad \text{so} \quad V = \sqrt{2196} = 6\sqrt{61} \approx 46.86.$$

The initial speed is $6\sqrt{61} \approx 46.86$ m s^{-1} and the angle of projection is about $50^\circ 12'$.

Chapter 10 Rates and Projectile Motion - Homework Answers

Lesson 10B | Projectile motion: the equation of path

Lesson Examples

C1 **Tutor demo. a** $t = \dfrac{x}{30}$, so $y = -\dfrac{x^2}{180} + \dfrac{4}{3}x$. **b** $y = 0$ gives the range $x = 240$ m; the greatest height is at $x = 120$: $y = 80$ m. **c** $\dfrac{dy}{dx} = -\dfrac{x}{90} + \dfrac{4}{3}$, which is $\frac{4}{3}$ at $x = 0$, so the angle of projection is $\tan^{-1}\frac{4}{3} \doteq 53°8'$.

Your turn. $y = -\dfrac{x^2}{80} + \dfrac{3}{4}x$. Range 60 m, greatest height 11.25 m (at $x = 30$), angle of projection $\tan^{-1}\frac{3}{4} \doteq 36°52'$.

C2 **Tutor demo. b** $45 = \dfrac{900}{10}\sin 2\alpha$, so $\sin 2\alpha = \frac{1}{2}$, $2\alpha = 30°$ or $150°$, and $\alpha = 15°$ or $75°$. The maximum range is $\dfrac{V^2}{g} = 90$ m.

Your turn. $\sin 2\alpha = \dfrac{\sqrt{3}}{2}$, so $\alpha = 30°$ or $60°$. Greatest heights $\dfrac{V^2 \sin^2 \alpha}{2g}$: 5 m and 15 m.

C3 **Tutor demo. a** $\dfrac{g}{2V^2} = \dfrac{10}{800} = \dfrac{1}{80}$. **b** $5 = 20T - 5(1 + T^2)$, where $T = \tan\alpha$, so $T^2 - 4T + 2 = 0$ and $T = 2 \pm \sqrt{2}$: $\alpha \doteq 74°$ or $30°$. **c** $10 = 40T - 20(1 + T^2)$ gives $2T^2 - 4T + 3 = 0$, with $\Delta = 16 - 24 < 0$: no real angle.

Your turn. $10 = 30T - 5(1 + T^2)$, so $T^2 - 6T + 3 = 0$, $T = 3 \pm \sqrt{6}$: $\alpha \doteq 80°$ or $29°$.

Textbook Homework

1 **a** 13.8 m **b** 18 m or 68.4 m **c** $\dfrac{dy}{dx} = -\frac{5}{162}x + \frac{4}{3}$

2 **b** (i) 192 m (ii) 20 m (iii) $22.6°$ (iv) $6°$ below the horizontal

3 **a** $\sqrt{10}$ **d** (i) $9.5°$ above the horizontal (ii) $9.5°$ below the horizontal

4 **a** $y = -\frac{1}{5}x^2$ **b** 10 m **c** $76°$ below the horizontal

5 **a** $\dot{x} = 12\sqrt{3}$, $\dot{y} = -10t + 12$, $x = 12t\sqrt{3}$, $y = -5t^2 + 12t$ **d** $D \doteq 58.7$

6 **b** (i) 13.5 m/s (ii) $71°$ below the horizontal

7 **b** $\theta \doteq 28.2°$ or $61.8°$

8 **b** $60°$

9 **b** $62°22'$ or $37°5'$

11 **a** $x = \frac{1}{2}Vt\sqrt{2}$, $y = -\frac{1}{2}gt^2 + \frac{1}{2}Vt\sqrt{2}$

12 **c** (ii) $R = 18$ metres

Chapter 10 Rates and Projectile Motion - Homework Answers

Chapter 10 Review | Chapter review exercise

1 **b** $x = 60t\cos 40°$, $y = -5t^2 + 60t\sin 40°$ **c** (i) $t \doteq 7.7$ seconds (ii) 354.5 m (iii) 74.4 m

2 **a** 33.75 m **b** 96.58 m **c** 41.23 m/s

3 **b** $x = 15t$, $y = -5t^2 + 20t$ **c** $15\sqrt{2}$ m

4 **a** $\dot{x} = 5\sqrt{2}$, $x = 5t\sqrt{2}$, $\dot{y} = -10t + 5\sqrt{2}$, $y = -5t^2 + 5t\sqrt{2}$ **b** range: 10 metres, maximum height: 2.5 metres when $x = 5$ **c** (i) 1.6 metres (ii) $\tan^{-1}\frac{3}{5}$ below the horizontal **d** (i) $x = 3$ (ii) $\tan^{-1}\frac{2}{5}$ above the horizontal

5 **a** $\underset{\sim}{v} = 24\underset{\sim}{i} + (18 - 10t)\underset{\sim}{j}$ **b** $\underset{\sim}{r} = (24t)\underset{\sim}{i} + (18t - 5t^2)\underset{\sim}{j}$ **c** (i) 30 m/s (ii) $96\underset{\sim}{i} - 8\underset{\sim}{j}$ (iii) $43.2\underset{\sim}{i} + 16.2\underset{\sim}{j}$

6 **a** $\underset{\sim}{v} = 12\sqrt{5}\,\underset{\sim}{i} + 24\underset{\sim}{j}$ **b** $36\sqrt{5}\,\underset{\sim}{i} + 27\underset{\sim}{j}$ **c** It is falling, because the height was 28 m after 2 seconds (or show that $\dot{y} = -6 < 0$).

7 **b** 11.25 m

8 **a** Initially $\dot{x} = \sqrt{5}$ and $\dot{y} = 2\sqrt{5}$. **b** $\dot{x} = \sqrt{5}$, $x = t\sqrt{5}$, $\dot{y} = -10t + 2\sqrt{5}$, $y = -5t^2 + 2t\sqrt{5}$ **d** 1 metre **e** 2 metres **f** $\dot{x} = \sqrt{5}$, $\dot{y} = -2\sqrt{5}$, $v = 5$ m/s, $\theta = -\tan^{-1}2$ **g** $y = 2x - x^2$

9 46 m/s

10 **b** $y = -\frac{4}{45}x^2 + \sqrt{3}\,x$ **c** (i) 15 m (ii) $\sqrt{3}$ seconds

11 **a** $-\frac{35}{24}$ cm/s **b** $-\frac{1}{96}$ radians per second

Chapter 10 Rates and Projectile Motion - Homework Answers

Lesson 10B | Projectile motion: the equation of path

Past-Paper Homework - Worked Solutions

PP 1 **Ascham School, 2020, Mathematics Extension 1 Trial HSC Examination, Q7**

When $x = 0$, $y = 4$, so the original height is 4. Put $y = 4$: $-3x^2 + 2x = 0$, that is, $x(2 - 3x) = 0$. So $x = 0$ (the start) or $x = \dfrac{2}{3}$. It travels $\dfrac{2}{3}$ unit horizontally before it is back at height 4. **Answer (B).**

PP 2 **Girraween High School, 2021, Trial Higher School Certificate Examination, Mathematics Extension 1, Q13(b)**

For the path to be a function of x we need $\cos\theta \neq 0$. From $x = Vt\cos\theta$, $t = \dfrac{x}{V\cos\theta}$. Substituting into y,

$$y = V\sin\theta \times \frac{x}{V\cos\theta} - \frac{1}{2}g\left(\frac{x}{V\cos\theta}\right)^2 = x\tan\theta - \frac{gx^2}{2V^2\cos^2\theta}.$$

Since $\dfrac{1}{\cos^2\theta} = \sec^2\theta = 1 + \tan^2\theta$, the cartesian equation is

$$y = x\tan\theta - \frac{gx^2}{2V^2}\left(1 + \tan^2\theta\right).$$

PP 3 **North Sydney Girls High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q13(d)(iii)**

(iii) As in the given equation, write α for $\alpha°$. From $\underset{\sim}{r}$, $x = 30t\cos\alpha$ and $y = 30t\sin\alpha - 5t^2 + 20$. So $t = \dfrac{x}{30\cos\alpha}$, and

$$y = 30\sin\alpha \times \frac{x}{30\cos\alpha} - 5 \times \frac{x^2}{900\cos^2\alpha} + 20 = x\tan\alpha - \frac{x^2}{180}\sec^2\alpha + 20.$$

The target is at $x = 90$, $y = 1$. Using $\sec^2\alpha = 1 + \tan^2\alpha$,

$$1 = 90\tan\alpha - 45\left(1 + \tan^2\alpha\right) + 20, \quad \text{so} \quad 45\tan^2\alpha - 90\tan\alpha + 26 = 0.$$

By the quadratic formula,

$$\tan\alpha = \frac{90 \pm \sqrt{90^2 - 4\times 45\times 26}}{90} = \frac{90 \pm \sqrt{3420}}{90} = 1 \pm \frac{\sqrt{95}}{15}.$$

So $\tan\alpha \approx 0.3502$ or $\tan\alpha \approx 1.6498$, giving $\alpha \approx 19°18'$ or $\alpha \approx 58°47'$.

PP 4 **Sydney Grammar School, 2022, Trial Examination, Form VI Mathematics Extension 1, Q13(d)**

(i) The projectile lands when $y = 0$: $t\,(V\sin\alpha - 5t) = 0$. Since $t \neq 0$, the time of flight is $t = \dfrac{V\sin\alpha}{5}$. Substituting into x,

$$R = V\cos\alpha \times \frac{V\sin\alpha}{5} = \frac{V^2\sin\alpha\cos\alpha}{5} = \frac{V^2\sin 2\alpha}{10}.$$

(ii) For a given V, R is largest when $\sin 2\alpha$ is largest. Since $0° < \alpha < 90°$, we have $0° < 2\alpha < 180°$, and $\sin 2\alpha \leq 1$ with equality only when $2\alpha = 90°$. So $R_{\max}$ occurs when $\alpha = 45°$, and

$$R_{\max} = \frac{V^2 \times 1}{10} = \frac{V^2}{10}.$$

(iii) The target is 12 km $= 12\,000$ m away, so it must be within the maximum range: $\dfrac{V^2}{10} \geq 12\,000$. Then $V^2 \geq 120\,000$, so

$$V \geq \sqrt{120\,000} = 200\sqrt{3} \approx 346.4 > 343.$$

So the launch speed must be faster than the speed of sound, 343 m/s.

PP 5 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q14(c)**

(i) Since $\theta \neq 90^\circ$, $\cos\theta \neq 0$. From $\underset{\sim}{r}(t)$, $x = Vt\cos\theta$, so $t = \dfrac{x}{V\cos\theta}$. Substituting into $y = Vt\sin\theta - \frac{1}{2}gt^2$,

$$y = x\tan\theta - \frac{gx^2}{2V^2\cos^2\theta} = x\tan\theta - \frac{x^2}{4h}\left(1+\tan^2\theta\right),$$

because $h = \dfrac{V^2}{2g}$ gives $\dfrac{g}{2V^2} = \dfrac{1}{4h}$, and $\dfrac{1}{\cos^2\theta} = \sec^2\theta = 1+\tan^2\theta$.

(ii) A ball at angle θ passes through (a,b) when $b = a\tan\theta - \dfrac{a^2}{4h}\left(1+\tan^2\theta\right)$.

Multiplying by $4h$ and rearranging,

$$a^2\tan^2\theta - 4ha\tan\theta + \left(a^2+4hb\right) = 0.$$

This is a quadratic in $\tan\theta$. The two balls have different angles of elevation, so $\tan\theta_1$ and $\tan\theta_2$ must be two different real roots. Hence the discriminant is positive:

$$16h^2a^2 - 4a^2\left(a^2+4hb\right) > 0, \quad \text{that is,} \quad 4a^2\left(4h^2-4hb-a^2\right) > 0.$$

Since $a^2 > 0$, this gives $a^2 < 4h^2 - 4hb = 4h(h-b)$.

(iii) By the product of the roots of the quadratic in (ii),

$$\tan\theta_1\tan\theta_2 = \frac{a^2+4hb}{a^2} = 1 + \frac{4hb}{a^2} \geq 1,$$

since $h > 0$ and (a,b) is not below the ground, so $b \geq 0$. But if $0^\circ < \theta_1 < 45^\circ$ and $0^\circ < \theta_2 < 45^\circ$, then $0 < \tan\theta_1 < 1$ and $0 < \tan\theta_2 < 1$, so $\tan\theta_1\tan\theta_2 < 1$. This is a contradiction. So two balls with $\theta_1 < 45^\circ$ and $\theta_2 < 45^\circ$ cannot both pass through (a,b).